# Sample Research Article

A derivation of the PreTeXt Sample Article

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**Abstract**

This is a sample of many of the things you can do with PreTeXt. Sometimes the math makes sense, sometimes it seems to be written in the first person, sort of like this Abstract.

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**2010 Math Subject Classification**. Primary 00-01, 00-02; Secondary 00A99.

## 1 The Fundamental Theorem

There is a remarkable theorem:[1]

**Theorem 1.1 The Fundamental Theorem of Calculus.** *If $f(x)$ is continuous, and the derivative of $F(x)$ is $f(x)$, then*

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

*Proof.* Left to the reader. ■

You will find almost nothing about all this in the article [2], nor in the book [1], since they belong in some other article, but we can cite them out-of-order for practice anyway.

When we are writing we do not always know what we want to cite, or just where subsequent material will end up. For example, we might want a citation to `[provisional cross-reference: some textbook about the FTC]`

This is an example of a statement describing funding support for the current document.
[1] And fortunately we do not need to try to write it in the margin!

or we might want to reference a later `[provisional cross-reference: chapter about DiffEq's, and an_underscore]`.

We can also embed "todo"s in the source by making an XML comment that begins with the four characters `todo`, and selectively display them, so you may not see the one here in the output you are looking at now. Or maybe you do see it?

Because a definite integral can be computed using an antiderivative, we have the following definition.

**Definition 1.2** Suppose that $\frac{d}{dx}F(x) = f(x)$. Then the **indefinite integral** of $f(x)$ is $F(x)$ and is written as

$$\int f(x)\,dx = F(x).$$

◊

# 2 Mathematics

To be able to create both LaTeX and HTML output (plus variations), we rely on MathJax, which in turn supports an extensive subset of the mathematical symbols normally available. The AMSMath symbol set is a good approximation. The PreTeXt Guide has a link to the complete list of macros supported by MathJax. We load the `AMSsymbols` library.

## 2.1 Basic Mathematics

The following is from the MathJax demonstration page, an identity due to Ramanujan:

$$\frac{1}{\Bigl(\sqrt{\phi\sqrt{5}}-\phi\Bigr)e^{\frac{2}{5}\pi}} = 1+\frac{e^{-2\pi}}{1+\frac{e^{-4\pi}}{1+\frac{e^{-6\pi}}{1+\frac{e^{-8\pi}}{1+\ldots}}}}$$

And again, from the MathJax demonstration page, Maxwell's equations:

$$\begin{aligned}
\nabla \times \vec{\mathbf{B}} -\, \frac1c\, \frac{\partial\vec{\mathbf{E}}}{\partial t} & = \frac{4\pi}{c}\vec{\mathbf{j}} \\
\nabla \cdot \vec{\mathbf{E}} & = 4 \pi \rho \\
\nabla \times \vec{\mathbf{E}}\, +\, \frac1c\, \frac{\partial\vec{\mathbf{B}}}{\partial t} & = \vec{\mathbf{0}} \\
\nabla \cdot \vec{\mathbf{B}} & = 0
\end{aligned}$$

Historically, we provided internal support for the LaTeX package `extpfeil`. As of 2023-10-19 this has become an author election (see the `<docinfo>` section in the source of this document). We preeserve a small test that this extensible arrows library is being included properly:

$$A \xmapsto[\text{bijection}]{\Phi+\Psi+\Theta} B$$

Look back at the top of the source file of this document to see how to include your TeX macros just once. For best results keep your macros simple and semantic.

PreTeXt once provided modest built-in support for “slanted”, or “beveled”, or “nice” fractions. To wit, we mean fractions such as: $\sfrac{3}{8}$. Use the pre-defined `\sfrac{}{}` macro in your mathematics to achieve this presentation. The presentation in HTML is subpar, but could improve as MathJax provides support. It is now an author’s responsibility to add support for superior typesetting for PDF output by loading the `xfrac` LaTeX package with the following in `<docinfo>`:

```
<math-package latex-name="xfrac" mathjax-name=""/>
```

which is what we have done here as a test. See the Guide for more details.

We consider a system of equations. We number the first and last equation (there are just two) and include an `xml:id` on each. We reference the whole system later as the range of equations from the first to the last.

$$\frac{dx}{dt} = x^2 - 4x - y + 4 \tag{2.1}$$

$$\frac{dy}{dt} = x^3 - y. \tag{2.2}$$

## 2.2 Displayed Mathematics

Multi-line displays of mathematics are achieved with the `md` tag (“math display”), and the variant that produces numbers on each line, `mdn` (“math display numbered”), used within a paragraph (`p`). As a good example of how XML syntax is superior, you author $n$ lines of equations by enclosing each line inside of a `mrow` tag, rather than using $n-1$ separators (such as `\\`).

If you use no ampersands to express alignment (read ahead), then each equation is centered independently on the width of the text. This is implemented according to the AMSmath LaTeX package’s `gather` environment. Example:

$$\begin{gather} \frac{dx}{dt} = x^2 - 4x - y + 4 \\ \frac{dy}{dt} = x^3 - y. \end{gather}$$

An ampersand is used, in two ways, to describe positioning several equations per line, organized in columns. We have created the pre-defined LaTeX macro `\amp` as one way specify these, but the escape sequence `&` may be used also. The second, fourth, sixth, … ampersands separate columns, and the spacing between columns will be provided automatically. The first, third, fifth, … ampersands are alignment points for the equations in each column. Typically this is placed just prior to a binary operator, such as an equal sign (`\amp =` ), or for a column of explanations or commentary, just prior to the `\text{}` macro. Note that this scenario suggests always having an odd number of ampersands in each `mrow`. In the example below, alignment is on the equals sign in the first two columns, and provides left-justification to the explanations in the third column. N.B.: the use below of the `\text{}` macro does not include mathematics within its argument. Doing so may yield unpredictable results depending on your choice of delimiters for the mathematics (and using an `m` tag will be ineffective).

$$\begin{align*} \frac{dx}{dt} &= x^2 - 4x - y + 4 & \frac{dy}{dt} &= x^3 - y & &x, y\text{ version}\\ \frac{dw}{dt} &= z^3 - w & \frac{dz}{dt} &= z^2 - 4z - w + 4 & &z, w\text{ version} \end{align*}$$

PreTeXt will automatically detect the presence or absence of ampersands, but by defining macros for entire aligned equations, you can effectively hide the ampersands. So the `@alignment` attribute can override automatic detection. We use a simple LaTeX macro to demonstrate setting `alignment='align'` to override the use of a `gather` environment and use a `align` environment instead. Example:

$$\begin{aligned}\frac{dx}{dt} &= x^2 - 4x - y + 4\\ \frac{dy}{dt} &= x^3 - y.\end{aligned}$$

The AMSmath LaTeX package's `alignat` environment is a third variant of alignment. It never happens automatically, you need to ask for it with `alignment="alignat"`. It is very similar to `align` but adds no space between the equation columns. So you can leave it that way, or you can add your own "extra" space to suit. Here is a previous example with no inter-column space:

$$\begin{alignedat}{3}\frac{dx}{dt} &= x^2 - 4x - y + 4 & \frac{dy}{dt} &= x^3 - y &\qquad\qquad & x, y \text{ version}\\ \frac{dw}{dt} &= z^3 - w & \frac{dz}{dt} &= z^2 - 4z - w + 4 & & z, w \text{ version.}\end{alignedat}$$

This modified example has a middle row with three columns, while the other rows have just one column, as a test of our routines for determining the `mrow` with the greatest number of ampersands (and how many there are),

$$\begin{alignedat}{3}\frac{dw}{dt} &= z^3 - w\\ \frac{dx}{dt} &= x^2 - 4x - y + 4 & \frac{dy}{dt} &= x^3 - y & x, y &\text{ third column}\\ \frac{dw}{dt} &= z^3 - w.\end{alignedat}$$

Final example demonstrates that ampersands in other objects (matrices here) can wreak havoc with computing the number of columns. So we provide yet another attribute to override automatic detection, `alignat-columns`. This is the number of *columns* not the number of *ampersands*. Generally, for $c$ columns, there will be $2c - 1$ ampersands.

$$A = \begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix} I = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}.$$

One caveat: if your number of ampersands is even (see advice above about using an odd number) behavior should still be correct, as in next example.

If you want super-precise control over alignment of the terms of a system of equations (linear or not) you can use the `alignat` option to advantage by not including any extra space. This example is modified slightly from a post by Alex Jordan:

$$\begin{alignedat}{4}2x &+{} & y &+{} & 3z &={} & 10\\ x & & &+{} & z &={} & 6\\ x &+{} & 3y &+{} & 2z &={} & 13.\end{alignedat}$$

Beautiful.

A long equation, to check layout on various screen sizes. This is Weil's "explicit formula" for the Riemann $\zeta$-function:

$$\sum_{\gamma} S_{-}(\gamma) = \frac{\log Q}{\pi}\hat{S}_{-}(0) + \frac{1}{2\pi}\sum_{j=1}^{d}\Re\left\{\int_{-\infty}^{\infty}\frac{\Gamma'}{\Gamma}\left(\frac{1}{4}+\frac{it}{2}+\mu_j\right)S_{-}(t)dt\right\} - \frac{d}{2\pi}\hat{S}_{-}(0)\log\pi. \tag{2.3}$$

**Example 2.1 Excessive Display Mathematics.** In print versions, a long run of displayed equations often needs to be broken across pages. If you are reading some other version of this, then there is nothing to see here. But for LaTeX output it could be interesting. First, with no extra effort, this page-long display should break naturally, no matter how the preceding material changes.

$$x^2 + y^2 = z^2 \tag{2.4}$$
$$a^2 + b^2 = c^2 \tag{2.5}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.6}$$
$$m^2 + n^2 = p^2 \tag{2.7}$$
$$x^2 + y^2 = z^2 \tag{2.8}$$
$$a^2 + b^2 = c^2 \tag{2.9}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.10}$$
$$m^2 + n^2 = p^2 \tag{2.11}$$
$$x^2 + y^2 = z^2 \tag{2.12}$$
$$a^2 + b^2 = c^2 \tag{2.13}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.14}$$
$$m^2 + n^2 = p^2 \tag{2.15}$$
$$x^2 + y^2 = z^2 \tag{2.16}$$
$$a^2 + b^2 = c^2 \tag{2.17}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.18}$$
$$m^2 + n^2 = p^2 \tag{2.19}$$
$$x^2 + y^2 = z^2 \tag{2.20}$$
$$a^2 + b^2 = c^2 \tag{2.21}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.22}$$
$$m^2 + n^2 = p^2 \tag{2.23}$$
$$x^2 + y^2 = z^2 \tag{2.24}$$
$$a^2 + b^2 = c^2 \tag{2.25}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.26}$$
$$m^2 + n^2 = p^2 \tag{2.27}$$
$$x^2 + y^2 = z^2 \tag{2.28}$$
$$a^2 + b^2 = c^2 \tag{2.29}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.30}$$
$$m^2 + n^2 = p^2 \tag{2.31}$$
$$x^2 + y^2 = z^2 \tag{2.32}$$
$$a^2 + b^2 = c^2 \tag{2.33}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.34}$$

$$m^2 + n^2 = p^2 \tag{2.35}$$
$$x^2 + y^2 = z^2 \tag{2.36}$$
$$a^2 + b^2 = c^2 \tag{2.37}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.38}$$
$$m^2 + n^2 = p^2 \tag{2.39}$$
$$x^2 + y^2 = z^2 \tag{2.40}$$
$$a^2 + b^2 = c^2 \tag{2.41}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.42}$$
$$m^2 + n^2 = p^2 \tag{2.43}$$
$$x^2 + y^2 = z^2 \tag{2.44}$$
$$a^2 + b^2 = c^2 \tag{2.45}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.46}$$
$$m^2 + n^2 = p^2 \tag{2.47}$$
$$x^2 + y^2 = z^2 \tag{2.48}$$
$$a^2 + b^2 = c^2 \tag{2.49}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.50}$$
$$m^2 + n^2 = p^2. \tag{2.51}$$

In this version we have turned off page breaking for the entire display, but then allowed a break at every fourth equation, so you should see a reasonably attractive page break right after one of the $m^2 + n^2 = p^2$ equations.

$$x^2 + y^2 = z^2 \tag{2.52}$$
$$a^2 + b^2 = c^2 \tag{2.53}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.54}$$
$$m^2 + n^2 = p^2 \tag{2.55}$$
$$x^2 + y^2 = z^2 \tag{2.56}$$
$$a^2 + b^2 = c^2 \tag{2.57}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.58}$$
$$m^2 + n^2 = p^2 \tag{2.59}$$
$$x^2 + y^2 = z^2 \tag{2.60}$$
$$a^2 + b^2 = c^2 \tag{2.61}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.62}$$
$$m^2 + n^2 = p^2 \tag{2.63}$$
$$x^2 + y^2 = z^2 \tag{2.64}$$
$$a^2 + b^2 = c^2 \tag{2.65}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.66}$$
$$m^2 + n^2 = p^2 \tag{2.67}$$
$$x^2 + y^2 = z^2 \tag{2.68}$$
$$a^2 + b^2 = c^2 \tag{2.69}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.70}$$
$$m^2 + n^2 = p^2 \tag{2.71}$$

$$x^2 + y^2 = z^2 \tag{2.72}$$
$$a^2 + b^2 = c^2 \tag{2.73}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.74}$$
$$m^2 + n^2 = p^2 \tag{2.75}$$
$$x^2 + y^2 = z^2 \tag{2.76}$$
$$a^2 + b^2 = c^2 \tag{2.77}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.78}$$
$$m^2 + n^2 = p^2 \tag{2.79}$$
$$x^2 + y^2 = z^2 \tag{2.80}$$
$$a^2 + b^2 = c^2 \tag{2.81}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.82}$$
$$m^2 + n^2 = p^2 \tag{2.83}$$
$$x^2 + y^2 = z^2 \tag{2.84}$$
$$a^2 + b^2 = c^2 \tag{2.85}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.86}$$
$$m^2 + n^2 = p^2 \tag{2.87}$$
$$x^2 + y^2 = z^2 \tag{2.88}$$
$$a^2 + b^2 = c^2 \tag{2.89}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.90}$$
$$m^2 + n^2 = p^2 \tag{2.91}$$
$$x^2 + y^2 = z^2 \tag{2.92}$$
$$a^2 + b^2 = c^2 \tag{2.93}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.94}$$
$$m^2 + n^2 = p^2 \tag{2.95}$$
$$x^2 + y^2 = z^2 \tag{2.96}$$
$$a^2 + b^2 = c^2 \tag{2.97}$$
$$\alpha^2 + \beta^2 = \gamma^2 \tag{2.98}$$
$$m^2 + n^2 = p^2. \tag{2.99}$$

So. Do not take any extra steps and let LaTeX figure out the breaks. If you do not like a break, modify the `md` or `mdn` to go back to the AMSmath default behavior and not break at all. Ever. Or rather, go further and modify an individual `mrow` to suggest that it is a good place for a break. □

This is a poorly-authored paragaph to test the conversion to HTML. There are two displayed equations, separated by a period ending the first one's sentence, which should migrate into the display, and not leave behind an empty paragraph:

$$z + y = z.$$

$$a + b = c.$$

This final sentence should remain, inside another HTML paragraph, without the second equation's period.

## 2.3 LaTeX Packages and MathJax Extensions

If you would like to enhance your mathematics by using a macro from a LaTeX package *and* there is a MathJax extension *which implements the same macro*, then you may use this with your mathematics as we demonstrate here.

This example is from Jason Underdown. The package is named `cancel` and is included in the TeXLive distribution, so is fairly standard. The particular macro being demonstrated is `\cancelto{}{}`.

$$\lim_{b\to\infty}\left[-\frac{1}{s}\cancelto{0}{e^{-sb}}+\frac{1}{s}\right].$$

Look at the source of this article to see the package name being supplied in a `<math-package>` element within the `<docinfo>` section. That is the only setup required to make the macro usable in LaTeX and HTML output.

See the *PreTeXt Guide* subsection about “MathJax Extensions” for more detail.

## 2.4 Advanced Mathematics

MathJax is extremely capable in rendering a subset of LaTeX in web browsers, and improving all the time. You can get fairly fancy with some of its supported commands. In particular, if you need to mix in a few words with your mathematics, the `\text{}` macro is supported. For example, you might use an “if” or an “otherwise” in the definition of a piecewise function.

Consider that the first line below is text sandwiched in-between two Greek letters, wrapped in a `\text{}` macro. In HTML output we have taken care that the font for text material within display mathematics should match the font of the surrounding paragraph, as also happens with LaTeX output. The second line is nearly identical in the source, but is just naked text being rendered like a slew of variables.

$$\begin{gathered}\alpha\text{ is not equal to }\beta\\ \alpha is not equal to \beta\\ \alpha\neq\beta.\end{gathered}$$

We are not suggesting here that using words in place of symbols, as in the first line, is a good practice. (It is not.)

The following example is a good stress-test of using the `\text{}` macro to achieve certain effects. Note the Unicode left and right smart quotes. This a contribution from Alex Jordan as part of his work on *APEX Calculus*.

$$y\to\frac{\sin(0)}{0}\to\text{“}\frac{0}{0}\text{”}.$$

And another one from Alex. Note the use of the `\mathord{}` and `\mathrel{}` macros to control spacing around the mathematical symbols. Examine the source to see how the quotation marks have been authored with XML syntax for Unicode characters, since we do not allow most markup inside mathematics.

$$\zeta(1)=\sum_{n=1}^{\infty}\frac{1}{n}\mathrel{\mathord{“}=\mathord{”}}\prod_{p}\left(\frac{1}{1-1/p}\right)=\prod_{p}\left(\frac{1}{1-p^{-1}}\right)$$

Scott Beaver likes to write short chains of equalities all in one line, with the cross-references sitting on each equals sign. Here we test the LaTeX `\overset` and \underset macros wrapping a PreTeXt <xref>, with and without content, inside an <me> element. Note that \stackrel is obsolete, and \overunderset is not yet supported by MathJax (but see GitHub #2704). The mathematics is Scott's, the reasons are totally unrelated to the math.

$$AC - AD \overset{1.1}{=} A(C - D) \underset{1.2}{=} A0_{n\times p} \overset{\text{Thm. 1.1}}{=} 0_{m\times p}$$

We suggest using cross-references that only display numbers (<xref> with @text set to global) since if you stick to elements like <theorem>, <lemma>, <definition>, or <axiom>, then the numbers will be unambiguous and the target of the cross-reference will contain full information. But note that if you mix in divisions, or perhaps figures, as reasons then there is a possibility that numbers will need to be qualified by their type. We have provided an abbreviation for one cross-reference to Theorem 1.1 (which will not benefit from automatic translation to other languages).

# A Multiple References

## A.1 Multiple Specialized References

You might want to have lists of references, in the back, but with multiple such lists. Make an <appendix> to hold them, give it some structure (for an <article>, a leading <subsection>, such as the one you are reading right now), then follow with multiple <references> divisions. A typical citation will then look like: [A.3.2].

## A.2 General References


[**1**] Gilbert Strang, *The Fundamental Theorem of Linear Algebra*, The American Mathematical Monthly November 1993, **100** no. 9, 848–855.


## A.3 Specialized References


[**1**] Gilbert Strang, *The Fundamental Theorem of Linear Algebra*, The American Mathematical Monthly November 1993, **100** no. 9, 848–855.

[**2**] Gilbert Strang, *The Fundamental Theorem of Linear Algebra*, The American Mathematical Monthly November 1993, **100** no. 9, 848–855.


# References


[**1**] Tom Judson, *Abstract Algebra: Theory and Applications*.
Another online, open-source offering.

[**2**] David C. Lay, *Subspaces and Echelon Forms*. The College Mathematics Journal, January 1993, **24** no. 1, 57–62.


### Colophon


This article was authored in, and produced with, PreTeXt. It is typeset with the Latin Modern font.